

Given enough Capacity, a neural network can fit training data perfectly. However, it does not mean that it can **generalize** well to testing data set.

Generalization error can be decomposed into three parts

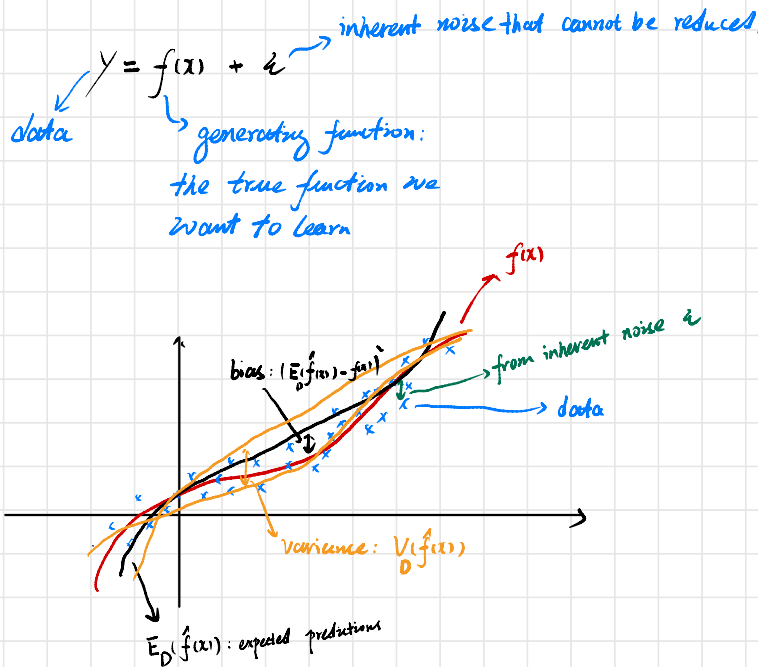
1. **inherent noise**: this comes from the inherent noise from the data.

In other words, even if we fit the generating function of the data perfectly, there is error that cannot be reduced.

2. **bias**: the difference between the expected predictions and the underlying real value.

3. **Variance**: the Variance of predictions given different training.

Illustration:



Derivations:

$$\begin{aligned}\bar{E}_{D, \epsilon}(\hat{f}(x; D) - y)^2 &= \bar{E}_{D, \epsilon}(\hat{f}(x; D) - f(x) - \epsilon)^2 \\&= \bar{E}_{D, \epsilon}((\hat{f}(x; D) - f(x))^2 + \epsilon^2 + 2\epsilon(\hat{f}(x; D) - f(x))) \\&= \bar{E}_{D, \epsilon}((\hat{f}(x; D) - f(x))^2) + E(\epsilon^2) + 0 \\&= \bar{E}_{D, \epsilon}((\hat{f}(x; D) - E_D(\hat{f}(x; D)) + E_D(\hat{f}(x; D)) - f(x))^2) + 0^2 \\&\quad \text{drop } \epsilon \\&= E_D((\hat{f}(x; D) - E_D(\hat{f}(x; D)))^2) + E_D((E_D(\hat{f}(x; D)) - f(x))^2) \\&\quad + E_D(2(\hat{f}(x; D) - E_D(\hat{f}(x; D)))(E_D(\hat{f}(x; D)) - f(x))) + 0^2 \\&= \text{Var}_D(\hat{f}(x; D)) + (f(x) - E_D(\hat{f}(x; D)))^2 + 0^2 \\&\quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \\&\quad \text{Variance} \quad \quad \text{bias} \quad \quad \text{inherent noise}\end{aligned}$$

Question: given enough data (as many as we want), is it possible to decompose model's prediction error into variance, bias and inherent noise?

Variance is easy to estimate, simply sample randomly the training data, fit the model and then estimate the variance.

the hard part is to decompose the bias and the inherent noise.

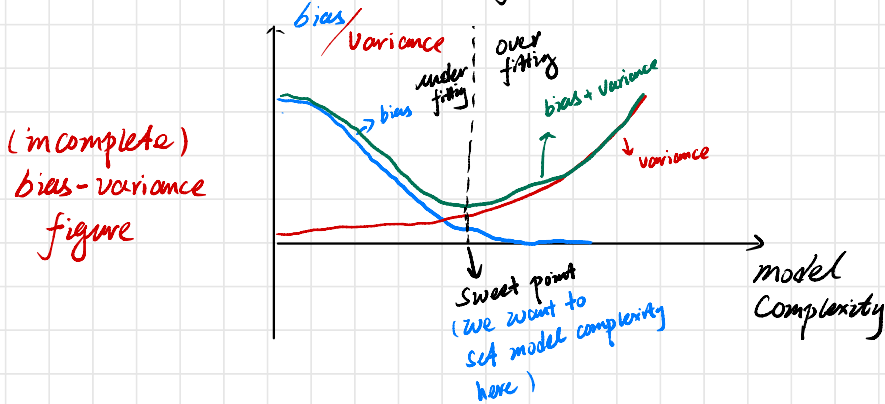
trick: Consider a model always have 0 bias,
(this model would always predict y , the data itself)
So we are left with variance and inherent noise

can estimate this now

Implication for model training.

The inherent noise can not be reduced, so we want to find a model with low bias and low variance.

However, this is hard because the famous bias-variance trade off.



As shown in the figure, we cannot simply increase the model complexity to very high and then expect a good result. The key is to strike the balance between bias and variance.

Common methods

To decrease bias: increase model complexity,
or choose a correct model
(e.g. a linear model can not fit an exponential function)

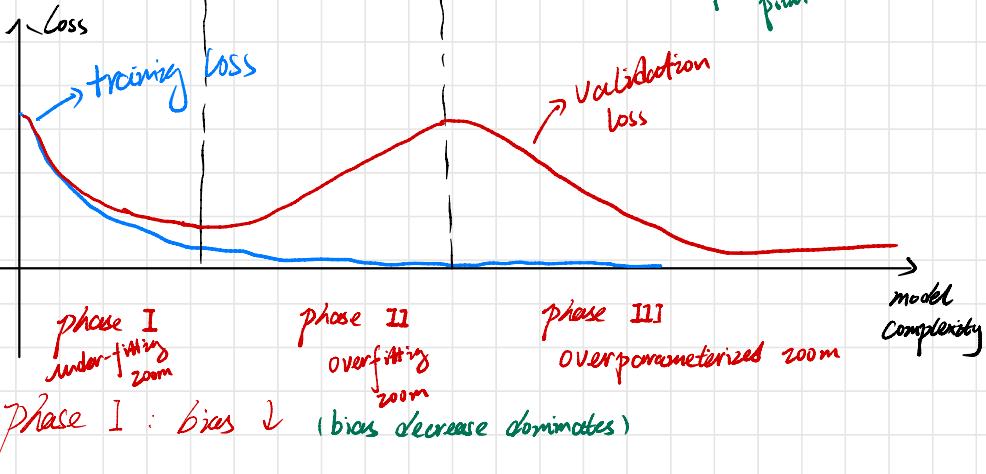
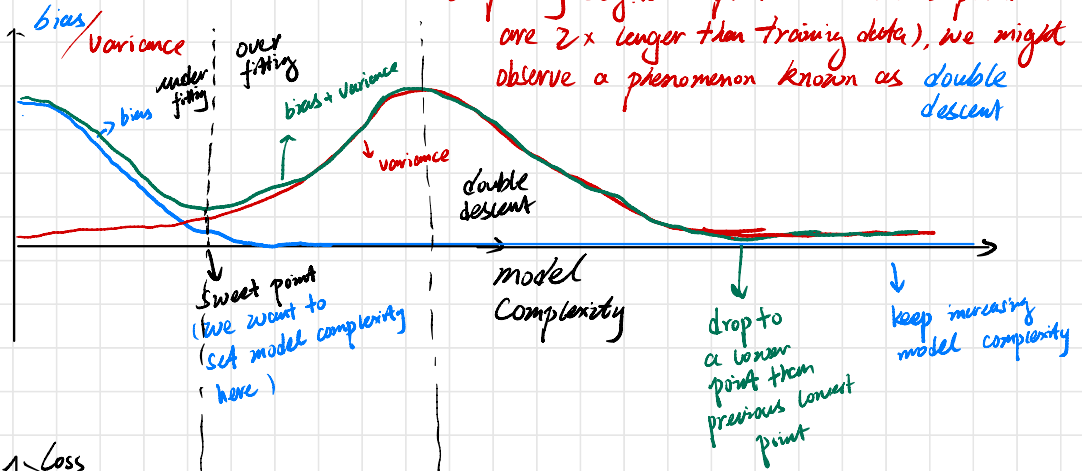
To decrease variance: 1. increase training data size

2. decrease model complexity

3. consider ensemble method. (e.g. random forest, keep or even increase bias, but decrease variance)

4. add regularization

It turns out the bias-variance figure is incomplete. if we keep increasing the model complexity (e.g. to the point where model parameters are 2x longer than training data), we might observe a phenomenon known as double descent



Phase I: bias ↓ (bias decrease dominates)

Phase II: Variance ↑ (variance increase dominates)

Phase III: Variance ↓ (variance decrease dominates)
 why? One explanation:
 when parameters > training samples,
 model has passed (interpolated) all instances, so
 all changes have to happen between training points.

In other words, after training error is 0, the model now

has the freedom to choose smooth interpretation between training points

possibly forced by
 all the explicit/implicit regularization technique
 e.g. SGD, dropout, L_2 norm etc.

raise a new question:
 why does the model choose to
 smooth the path (when train error is already 0)

Note: the model is still over-fitting (see the gap between train and validation data), but the model generalize better